



STUDENT INNOVATIVE PROJECT 2025- 2026 COMPLETION REPORT



DEVELOPMENT OF A HYBRID FIXED WING DRONE(VTOL)

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Certified that this report titled as, “**DEVELOPMENT OF A HYBRID FIXED WINGDRONE(VTOL)**” is the Bonafide work of **HARISHWAR A (2023105012)**, **Vishal R (2023105023)** and **Surya M (2023105527)** who carried out the project work for **CSRC**, in the month of may, 2026 under my supervision. Certified further that to the best of my knowledge, the work reported herein does not form part of any other thesis or dissertation based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

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I thank all the teaching and non-teaching staff of the Department of Electronics and Communication Engineering, for their kind help and co-operation during the course of my project.

ABSTRACT

This project presents the development of a Fixed Wing Unmanned Aerial Vehicle (UAV) tailored for surveillance operations. Fixed-wing drones are well-suited for long-range missions due to their aerodynamic efficiency, higher flight speed, and extended endurance, compared to multirotor systems. These characteristics make them ideal for covering large areas in applications such as border monitoring, agricultural inspection, wildlife tracking, disaster assessment, and security surveillance. The proposed fixed-wing drone is built using lightweight materials to optimize flight time and stability. It is equipped with essential avionics including a , flight controller, and telemetry module. The drone supports both manual and autonomous flight modes, enabling flexible deployment in varied environments.

The fixed-wing configuration allows the drone to maintain stable, efficient flight paths, making it capable of extended missions with minimal power consumption. This project demonstrates the effectiveness of fixed-wing UAVs in delivering reliable, high-performance surveillance in both civilian and strategic applications.

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1.INTRODUCTION

1.1 PROJECT MOTIVATION

The increasing demand for real-time monitoring and data collection in remote or inaccessible areas has highlighted the limitations of traditional surveillance methods. Ground-based systems and manned aerial patrols are often costly, time-consuming, and risky, especially in hostile or disaster-prone environments. To address these challenges, Unmanned Aerial Vehicles (UAVs) have emerged as a transformative solution, offering flexibility, safety, and cost-effectiveness.

Among various UAV configurations, fixed-wing drones stand out for their long flight endurance, higher speeds, and the ability to cover vast areas efficiently. These features make them particularly suitable for extended surveillance missions, such as border security, agricultural monitoring, disaster assessment, and wildlife protection. Unlike multirotor drones, fixed-wing UAVs can operate for longer durations on a single charge, making them ideal for time-critical and wide-area operations.

The motivation behind this project is to leverage the advantages of fixed-wing drone technology to create a reliable, autonomous, and affordable surveillance platform. By integrating navigation systems, real-time imaging, and lightweight materials, this drone aims to support a wide range of surveillance applications while minimizing human risk and operational costs. The project also serves as a practical learning experience in aerodynamics, embedded systems, and UAV integration for students and enthusiasts alike.

1.2 PROJECT OVERVIEW

This fixed wing drone is built by using foam board (5mm thickness). The fixed wing VTOL (Vertical Take-Off and Landing) model can be built by using PLA material through 3D printing technology and foam board for light weight. For the fixed wing VTOL model, the SpeedyBee F405 Wing flight controller is used because it supports dual firmware flexibility, onboard sensors, and data logging. The VTOL model includes a 40A ESC (Electronic Speed Controller) and a 1200 KV brushless motor, controlled via Mission Planner.

The SpeedyBee F405 Wing FC has inbuilt sensors including a barometer and gyroscope. The barometer is used for estimation of altitude, and the gyroscope ensures stabilization, levelling, and motion tracking.

1.3 PROBLEM STATEMENT

Traditional surveillance methods, such as manned patrolling and fixed camera systems, face significant limitations in terms of cost, coverage area, accessibility, and real-time adaptability. In large or remote areas, such as international borders, disaster-affected zones, forests, or agricultural lands, it becomes challenging to monitor activities continuously and effectively using conventional means.

Multirotor drones, though capable of vertical take-off and hovering, suffer from limited flight time and range due to high energy consumption. This limits their usefulness in long-duration or wide-area surveillance missions. There is a need for a more efficient, reliable, and autonomous aerial platform that can cover large distances, operate with minimal ground infrastructure, and provide real-time data for decision-making.

The problem, therefore, is to design and develop a fixed-wing unmanned aerial vehicle (UAV) that can perform long-endurance surveillance missions, equipped with essential navigation and imaging systems, while being cost-effective, lightweight, and suitable for autonomous operations in varied environments.

1.4 OBJECTIVE

The primary objective of this project is to design and develop a fixed-wing unmanned aerial vehicle (UAV) capable of conducting efficient and reliable aerial surveillance over large areas. The drone should offer extended flight duration, autonomous navigation, and real-time data transmission for monitoring and reconnaissance purposes.

1. Designing an aerodynamic and lightweight fixed-wing airframe suitable for stable, long-endurance flight.
2. Implementing a flight controller (SpeedyBee F405 Wing FC) to manage stable flight control, altitude hold, and autonomous features.
3. Testing and validating the drone's performance under different environmental and operational conditions.

1.5 SUMMARY

This project involves the design and development of a Fixed-Wing Unmanned Aerial Vehicle (UAV) optimized for aerial surveillance applications. Fixed-wing drones offer superior flight efficiency, longer range, and greater endurance compared to multirotor systems, making them ideal for monitoring large or remote areas such as borders, forests, disaster zones, or agricultural lands.

The drone is built using a lightweight and aerodynamic airframe to ensure stable and energy-efficient flight. A modern flight controller (SpeedyBee F405 Wing FC) is integrated to manage flight operations, with support for GPS navigation, autonomous waypoint control, and real-time telemetry.

2.DESIGN OF FIXED WING DRONE

2.1 HARDWARE COMPONENTS

The SpeedyBee F405 Wing Flight Controller is a high-performance, feature-rich flight control system specifically designed for fixed-wing and VTOL drones. Built around the powerful STM32F405 microcontroller, it offers fast processing and stable performance required for autonomous and long-endurance missions. It includes essential sensors such as a BMP280 barometer for altitude measurement and an IMU (gyroscope and accelerometer) for precise flight stabilization. One of its standout features is built-in Bluetooth and Wi-Fi, allowing users to configure the drone wirelessly via the SpeedyBee mobile app, eliminating the need for USB connection in the field. With support for multiple UARTs, it can interface with GPS modules, telemetry radios, and airspeed sensors simultaneously. The flight controller is compatible with popular open-source firmware like INAV and ArduPilot, enabling advanced flight modes such as waypoint navigation, return-to-home (RTH), and loiter. Its versatility, ease of use, and robust capabilities make the SpeedyBee F405 Wing FC an ideal choice for fixed-wing UAV surveillance applications.



Fig.2.1 SpeedyBee F405 Wing

Table 2.1 Specifications of SpeedyBee F405 Wing FC

Category	Specification
Processor (MCU)	STM32F405, 32-bit ARM Cortex-M4, 168 MHz
IMU (Gyro & Accel)	ICM-42688-P or MPU6000 (depends on batch)
Barometer	BMP280 (for altitude sensing)
Flash Memory	16MB onboard Blackbox data logging
UART Ports	6 total – for GPS, telemetry, RC receiver, airspeed, etc.
I2C Port	1 I2C port – for external sensors like airspeed sensor
PWM Outputs	9 outputs – for ESCs, servos, and VTOL motors
Input Voltage	7V – 35V (Supports 2S to 8S LiPo)
BEC Output	5V/3A and 9V/2A for powering external devices
Wireless Configuration	Built-in Wi-Fi and Bluetooth (via SpeedyBee App – Android/iOS)
Firmware Compatibility	INAV, ArduPilot
Telemetry Protocols	FrSky SmartPort, FPort, SBUS, CRSF, ELRS, MAVLink
OSD Support	Yes – analog OSD supported
Current Sensor	Integrated analog sensor (supports up to 100A)
Mounting Dimensions	30.5mm × 30.5mm (standard M3 hole spacing)
Board Dimensions	~37mm × 37mm (excluding connectors)
Weight	Approx. 10 grams

The SpeedyBee F405 Wing flight controller is a multilayer printed circuit board. It is a 3-layer printed circuit board consisting of an FC board, Power Distribution Board (PDB), and wireless board.



Fig.2.2 FC Board, Power Distribution Board, Wireless Board

The 1200KV Brushless Motor is a high-performance, cost-effective motor designed for mid- to large-size drones and fixed-wing aircraft. With its 1200KV rating, it offers an excellent balance between thrust and efficiency, making it suitable for both freestyle and long-endurance flight applications. Built with high-quality materials, including Japanese NMB bearings and a precision-machined aluminum bell. Its stator size provides increased torque and stability, making it ideal for payload-carrying drones or fixed-wing platforms requiring strong propulsion.



Fig.2.3 1200KV Brushless Motor

Features:

4. Precision bearing
5. Steel shaft
6. Anodizing
7. 19×19mm hole pattern
8. Multi-strand winding
9. 200mm 18 AWG silicone wire
10. Fully mirror-polished finish that shines and glistens with even the smallest hint of light
11. Upgraded N52SH curved arc magnets for tighter airgap and better performance
12. Massive 12mm EZO bearing for unparalleled smooth operation

The 40A Electronic Speed Controller (ESC) is a high-performance and reliable component designed to handle demanding drone applications, including fixed-wing, multirotor, and VTOL platforms. It supports up to 5S LiPo batteries and provides a continuous current of 40 amps, making it suitable for powerful brushless motors used in aerial surveillance and long-endurance flights. Featuring BLHeli_32 firmware, the ESC offers smooth throttle response, programmable features, and high-speed signal processing for precise control. Its compact, robust design includes high-

quality MOSFETs, low-resistance PCB layout, and efficient heat dissipation to ensure durability and consistent performance even under high load. The 40A ESC is ideal for drone builders looking for a balance between performance, reliability, and ease of integration.



Fig.2.4 40 A Electronic Speed Controller (ESC)

Specifications:

- Firmware: BLHeli_S JH50
- Burst Current: 60A
- Power Input: 3S – 6S LiPo
- Power Output: VBAT

2.2 3D MODEL FOR VTOL

The image depicts a top-view CAD model of a hybrid VTOL (Vertical Take-Off and Landing) fixed-wing drone designed for aerial surveillance and long-endurance missions. The airframe combines the efficiency of a fixed-wing layout with the versatility of vertical lift. Two vertical lift motors are mounted on the wings to enable stable vertical take-off and landing, while a rear-mounted pusher propeller provides forward thrust during horizontal flight. The streamlined fuselage houses onboard electronics and payload, while the tail section features a V-tail configuration with control surfaces for improved aerodynamic control. This configuration allows the drone to operate in diverse environments, eliminating the need for a runway while maintaining extended flight duration and efficient cruising capabilities.

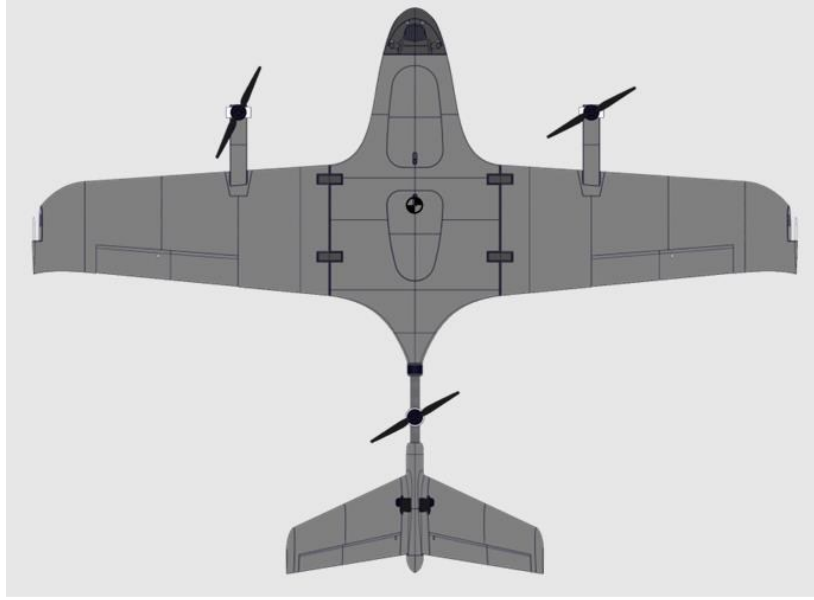


Fig.2.5 Sketch of the VTOL Fixed Wing Drone

The parts of the Fixed Wing VTOL drone are assembled using FreeCAD software. The VTOL fixed-wing drone model is designed with a focus on structural efficiency, printability, and flight performance. Its modular components include vertical lift motors on the wings and a rear-mounted pusher motor for forward thrust, enabling vertical take-off and efficient horizontal flight. To ensure lightweight yet durable construction, the drone incorporates a combination of materials based on functional requirements. PETG is used for high-stress structural parts such as the boom (left/right) and both front and tail motor mounts, offering strength and heat resistance. For the aerodynamic surfaces, including the three sets of VTOL wings (left/right), LW-PLA (Lightweight PLA) is used to significantly reduce the overall airframe weight while maintaining adequate rigidity. This combination of PETG and LW-PLA provides an optimal balance between performance, durability, and printability, making the drone suitable for long-endurance surveillance missions.



Fig.2.6 3D Model of the VTOL Drone

2.3 LAYOUT FOR RC FIXED WING

A Radio-Controlled (RC) Plane is a miniature aircraft that is operated remotely using a handheld radio transmitter. These aircraft are commonly powered by electric motors or internal combustion engines and are equipped with servos to control surfaces such as the rudder, elevator, and ailerons. RC planes are widely used in hobby flying, aeromodelling competitions, research, aerial photography, and even in educational and defense applications.

The core principle behind an RC plane is wireless communication between the transmitter and the onboard receiver, which interprets the pilot's inputs and actuates the corresponding control surfaces. RC planes can range from simple foam gliders for beginners to complex UAVs (Unmanned Aerial Vehicles) capable of autonomous navigation and long-range missions.

This project focuses on building an RC fixed-wing drone that combines the fundamentals of radio-controlled flight with modern electronics, sensors, and materials, making it suitable for real-world applications such as aerial surveillance, mapping, and environmental monitoring. The integration of flight controllers, GPS modules, and telemetry systems allows for enhanced stability, data collection, and semi-autonomous or fully autonomous operation, bridging the gap between hobby-grade and professional UAV platforms.

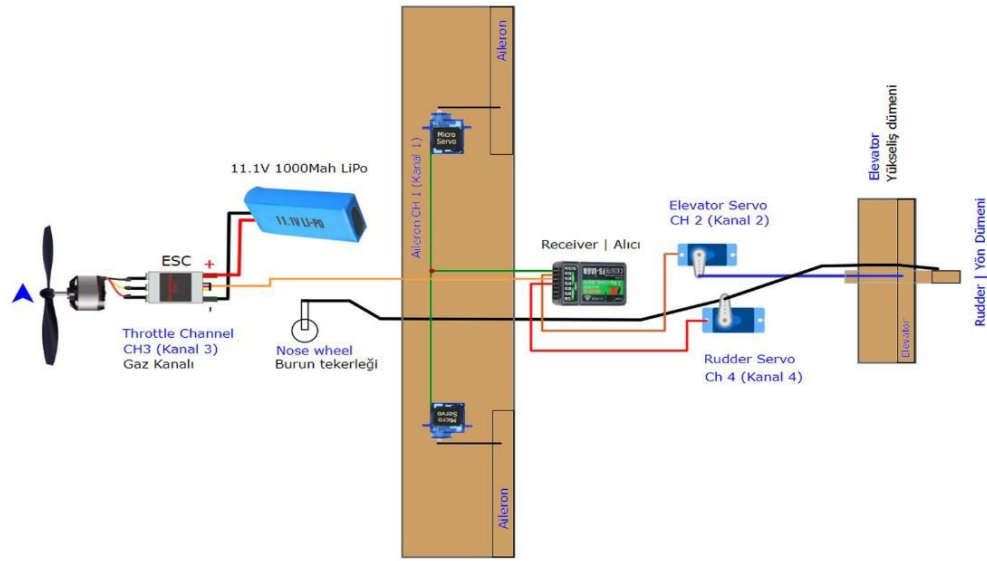


Fig.2.7 RC Plane Interior Setup

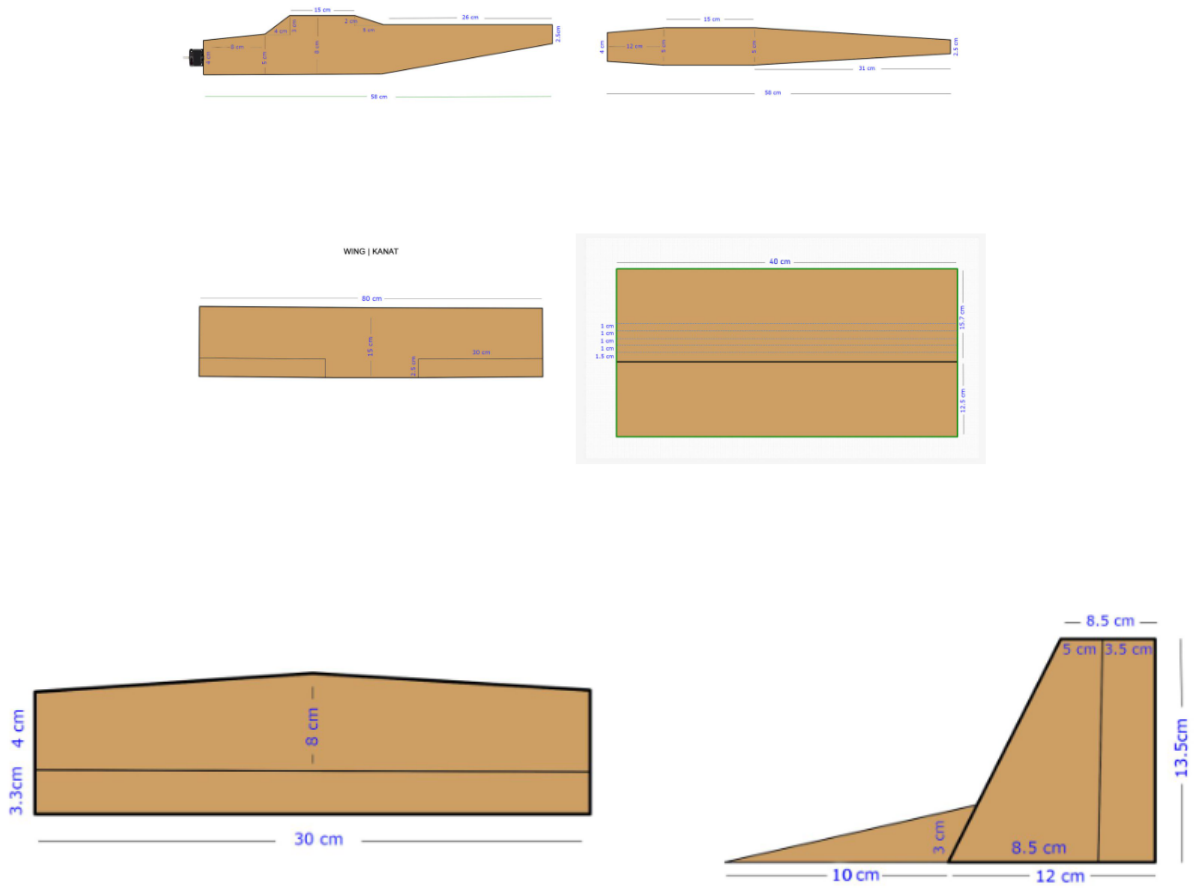


Fig.2.8 RC Plane Sketch

3.ELECTRONICS SETUP

3.1 FC CALIBRATION

FC calibration refers to the process of configuring and aligning the sensors within a Flight Controller (FC) so that the drone can fly accurately and stably. Since the flight controller is the “brain” of the drone, it relies on built-in sensors like the gyroscope, accelerometer, magnetometer (compass), and barometer to understand its orientation and movement. If these sensors are not calibrated properly, the drone may drift, tilt, or behave unpredictably.

Name	Δ	Value	Default	Units	Options	Desc	Fav
ARMING_SKIPCHK	786430	0				Checks to skip prior to arming motor. This is a bitmask of checks before allowing arming that will be skipped. For most users it is recommended to leave this at the default of 0 (no checks skipped). In extreme cases, users may wish to skip certain checks, such as the gyro or accelerometer, to allow for faster arming.	✓
BRD_SAFETY_DEFLT	0	0			0 Disabled 1 Enabled	This controls the default state of the safety switch at startup. When set to 1 the safety switch will start in the safe state (flaring) at boot. When set to zero the safety switch will start in the unsafe state (pulsed) at startup. Note that if a safety switch is fitted this value may still override this value, unless the safety switch is configured to be in the unsafe state at boot.	✓
COMPASS_USE	0	1			0 Disabled 1 Enabled	Enable or disable the use of the compass (instead of the GPS) for determining heading	✓
FLTMODE_CH	5	8			0 Disabled 1 Channel 1 2 Channel 2	RC Channel to use for flight mode control	✓
Q_A_RAT_YAW_D	0	0		°/2rad	0.000 0.02	Yaw axis rate controller D gain. Compensates for short-term change in desired yaw rate vs actual yaw rate	✓
Q_A_RAT_YAW_I	0.02	0.018	1/rad		0.010 1.0	Yaw axis rate controller I gain. Corrects long-term difference in desired yaw rate vs actual yaw rate	✓
Q_A_RAT_YAW_P	0.1	0.18	s/rad		0.10 2.50	Yaw axis rate controller P gain. Corrects in proportion to the difference between the desired yaw rate vs actual yaw rate	✓
Q_ASSIST_SPEED	10	0	m/s		0 100	This is the speed below which the quad motors will provide stability and lift assistance in fixed wing modes. The default value of 0 disables assistance but will generate a pre-arm failure to encourage users to set this parameter to a non-zero value. If this is set to a value of 0, ASSISTED mode will not be available.	✓
Q_ENABLE	1	0			0 Disable 1 Enable 2 Enable, VTOL, AUTO	This enables QuadPlane functionality, assuming multicopter motors start on output 5. If this is set to 2 then when starting AUTO mode it will initially be in VTOL AUTO mode.	✓
Q_FRAME_CLASS	7	1			0 Undefined 1 Quad 2 Hexa 3 Hexa+	Controls major frame class for multicopter component	✓
Q_FRAME_TYPE	1	1			0 Plan 1 X 2 Y 3 X+	Controls motor mixing for multicopter component	✓
Q_M_PWM_TYPE	0	0			0 Normal 1 OneShot 2 OneShot 10% 3 OneShot 10% 4 OneShot 10% 5 OneShot 10% 6 OneShot 10% 7 OneShot 10% 8 OneShot 10% 9 OneShot 10% 10 OneShot 10% 11 OneShot 10% 12 OneShot 10% 13 OneShot 10% 14 OneShot 10% 15 OneShot 10% 16 OneShot 10% 17 OneShot 10% 18 OneShot 10% 19 OneShot 10% 20 OneShot 10% 21 OneShot 10% 22 OneShot 10% 23 OneShot 10% 24 OneShot 10% 25 OneShot 10% 26 OneShot 10% 27 OneShot 10% 28 OneShot 10% 29 OneShot 10% 30 OneShot 10% 31 OneShot 10% 32 OneShot 10% 33 OneShot 10% 34 OneShot 10% 35 OneShot 10% 36 OneShot 10% 37 OneShot 10% 38 OneShot 10% 39 OneShot 10% 40 OneShot 10% 41 OneShot 10% 42 OneShot 10% 43 OneShot 10% 44 OneShot 10% 45 OneShot 10% 46 OneShot 10% 47 OneShot 10% 48 OneShot 10% 49 OneShot 10% 50 OneShot 10% 51 OneShot 10% 52 OneShot 10% 53 OneShot 10% 54 OneShot 10% 55 OneShot 10% 56 OneShot 10% 57 OneShot 10% 58 OneShot 10% 59 OneShot 10% 60 OneShot 10% 61 OneShot 10% 62 OneShot 10% 63 OneShot 10% 64 OneShot 10% 65 OneShot 10% 66 OneShot 10% 67 OneShot 10% 68 OneShot 10% 69 OneShot 10% 70 OneShot 10% 71 OneShot 10% 72 OneShot 10% 73 OneShot 10% 74 OneShot 10% 75 OneShot 10% 76 OneShot 10% 77 OneShot 10% 78 OneShot 10% 79 OneShot 10% 80 OneShot 10% 81 OneShot 10% 82 OneShot 10% 83 OneShot 10% 84 OneShot 10% 85 OneShot 10% 86 OneShot 10% 87 OneShot 10% 88 OneShot 10% 89 OneShot 10% 90 OneShot 10% 91 OneShot 10% 92 OneShot 10% 93 OneShot 10% 94 OneShot 10% 95 OneShot 10% 96 OneShot 10% 97 OneShot 10% 98 OneShot 10% 99 OneShot 10% 100 OneShot 10%	This selects the output PWM type, allowing for normal PWM continuous output, OneShot, brushed or DShot motor output. PWMRange and PWMAngle are PWM special cases for ESCs that don't support standard PWM. PWMRange is used for ESCs that don't support standard PWM. PWMAngle is used for ESCs that don't support standard PWM. PWMRange and PWMAngle are PWM special cases for ESCs that don't support standard PWM. PWMRange is used for ESCs that don't support standard PWM. PWMAngle is used for ESCs that don't support standard PWM.	✓
Q_M_SPIN_ARM	0.1	0.1			0.0 0.2 0.1 Low 0.2 High	Point at which the motors start to spin expressed as a number from 0 to 1 in the entire output range. Should be lower than MOT_SPIN_MIN.	✓
Q_M_SPIN_MIN	0.15	0.15			0.0 0.25 0.1 Low 0.2 High	Point at which the thrust starts expressed as a number from 0 to 1 in the entire output range. Should be higher than MOT_SPIN_ARM.	✓
Q_TILT_ENABLE	1	0			0 Disable 1 Enable	This enables Tiltrotor functionality	✓
Q_TILT_MASK	3	0				This is a bitmask of motors that are tiltable in a tiltrotor (or triwing). The mask is in terms of the standard motor order for the frame type.	✓
Q_TILT_MAX	45	45	deg		20 80	This is the maximum angle of the tiltable motors at which multicopter control will be enabled. Beyond this angle the plane will fly solely as a fixed wing aircraft and the motors will tilt to their maximum angle at the TILT_RATE.	✓
Q_TILT_RATE_ON	10	0	deg/s		10 300	This is the maximum speed at which the motor angle will change for a tiltrotor when moving from hover to forward flight. When this is zero the Q_TILT_RATE_UP value is used.	✓
Q_TILT_RATE_UP	10	40	deg/s		10 300	This is the maximum speed at which the motor angle will change for a tiltrotor when moving from forward flight to hover.	✓
Q_TILT_TYPE	0	0			0 Continuous 1 Binary	This is the type of tiltrotor when TILT_MASK is non-zero. A continuous tiltrotor can tilt the rotors to any angle on demand. A binary tiltrotor assumes a retract style servo where the servo is either fully forward or fully retracted. A binary tiltrotor will only move the rotors to 0 or 90 degrees.	✓
Q_TRANSITION_MS	5000	5000	ms		500 30000	Transition time in milliseconds after minimum airspeed is reached	✓
RCS_REVERSED	0	0			0 Normal 1 Reversed	Reverse channel input. Set to 0 for normal operation. Set to 1 to reverse this input channel.	✓
RCS_OPTION	153	0			0 Do Nothing 52 ACRO Mode 55 INVERT 56 INVERT 57 INVERT 58 INVERT 59 INVERT 60 INVERT 61 INVERT 62 INVERT 63 INVERT 64 INVERT 65 INVERT 66 INVERT 67 INVERT 68 INVERT 69 INVERT 70 INVERT 71 INVERT 72 INVERT 73 INVERT 74 INVERT 75 INVERT 76 INVERT 77 INVERT 78 INVERT 79 INVERT 80 INVERT 81 INVERT 82 INVERT 83 INVERT 84 INVERT 85 INVERT 86 INVERT 87 INVERT 88 INVERT 89 INVERT 90 INVERT 91 INVERT 92 INVERT 93 INVERT 94 INVERT 95 INVERT 96 INVERT 97 INVERT 98 INVERT 99 INVERT 100 INVERT	Function assigned to this RC channel	✓
SERVO_RATE	50	50	Hz		25 400	Default output rate in Hz for all PWM outputs	✓
SERVO1_FUNCTION	33	4			-1 GPIO 0 Disabled 1 GPIO 2 GPIO 3 GPIO 4 GPIO 5 GPIO 6 GPIO 7 GPIO 8 GPIO 9 GPIO 10 GPIO 11 GPIO 12 GPIO 13 GPIO 14 GPIO 15 GPIO 16 GPIO 17 GPIO 18 GPIO 19 GPIO 20 GPIO 21 GPIO 22 GPIO 23 GPIO 24 GPIO 25 GPIO 26 GPIO 27 GPIO 28 GPIO 29 GPIO 30 GPIO 31 GPIO 32 GPIO 33 GPIO 34 GPIO 35 GPIO 36 GPIO 37 GPIO 38 GPIO 39 GPIO 40 GPIO 41 GPIO 42 GPIO 43 GPIO 44 GPIO 45 GPIO 46 GPIO 47 GPIO 48 GPIO 49 GPIO 50 GPIO 51 GPIO 52 GPIO 53 GPIO 54 GPIO 55 GPIO 56 GPIO 57 GPIO 58 GPIO 59 GPIO 60 GPIO 61 GPIO 62 GPIO 63 GPIO 64 GPIO 65 GPIO 66 GPIO 67 GPIO 68 GPIO 69 GPIO 70 GPIO 71 GPIO 72 GPIO 73 GPIO 74 GPIO 75 GPIO 76 GPIO 77 GPIO 78 GPIO 79 GPIO 80 GPIO 81 GPIO 82 GPIO 83 GPIO 84 GPIO 85 GPIO 86 GPIO 87 GPIO 88 GPIO 89 GPIO 90 GPIO 91 GPIO 92 GPIO 93 GPIO 94 GPIO 95 GPIO 96 GPIO 97 GPIO 98 GPIO 99 GPIO 100 GPIO	Function assigned to this servo. Setting this to Disabled(0) will setup this output for control by auto missions or MAVLink servo set commands. any other value will enable the corresponding function	✓
SERVO2_FUNCTION	34	19			-1 GPIO 0 Disabled 1 GPIO 2 GPIO 3 GPIO 4 GPIO 5 GPIO 6 GPIO 7 GPIO 8 GPIO 9 GPIO 10 GPIO 11 GPIO 12 GPIO 13 GPIO 14 GPIO 15 GPIO 16 GPIO 17 GPIO 18 GPIO 19 GPIO 20 GPIO 21 GPIO 22 GPIO 23 GPIO 24 GPIO 25 GPIO 26 GPIO 27 GPIO 28 GPIO 29 GPIO 30 GPIO 31 GPIO 32 GPIO 33 GPIO 34 GPIO 35 GPIO 36 GPIO 37 GPIO 38 GPIO 39 GPIO 40 GPIO 41 GPIO 42 GPIO 43 GPIO 44 GPIO 45 GPIO 46 GPIO 47 GPIO 48 GPIO 49 GPIO 50 GPIO 51 GPIO 52 GPIO 53 GPIO 54 GPIO 55 GPIO 56 GPIO 57 GPIO 58 GPIO 59 GPIO 60 GPIO 61 GPIO 62 GPIO 63 GPIO 64 GPIO 65 GPIO 66 GPIO 67 GPIO 68 GPIO 69 GPIO 70 GPIO 71 GPIO 72 GPIO 73 GPIO 74 GPIO 75 GPIO 76 GPIO 77 GPIO 78 GPIO 79 GPIO 80 GPIO 81 GPIO 82 GPIO 83 GPIO 84 GPIO 85 GPIO 86 GPIO 87 GPIO 88 GPIO 89 GPIO 90 GPIO 91 GPIO 92 GPIO 93 GPIO 94 GPIO 95 GPIO 96 GPIO 97 GPIO 98 GPIO 99 GPIO 100 GPIO	Function assigned to this servo. Setting this to Disabled(0) will setup this output for control by auto missions or MAVLink servo set commands. any other value will enable the corresponding function	✓
SERVO4_FUNCTION	56	21			-1 GPIO 0 Disabled 1 GPIO 2 GPIO 3 GPIO 4 GPIO 5 GPIO 6 GPIO 7 GPIO 8 GPIO 9 GPIO 10 GPIO 11 GPIO 12 GPIO 13 GPIO 14 GPIO 15 GPIO 16 GPIO 17 GPIO 18 GPIO 19 GPIO 20 GPIO 21 GPIO 22 GPIO 23 GPIO 24 GPIO 25 GPIO 26 GPIO 27 GPIO 28 GPIO 29 GPIO 30 GPIO 31 GPIO 32 GPIO 33 GPIO 34 GPIO 35 GPIO 36 GPIO 37 GPIO 38 GPIO 39 GPIO 40 GPIO 41 GPIO 42 GPIO 43 GPIO 44 GPIO 45 GPIO 46 GPIO 47 GPIO 48 GPIO 49 GPIO 50 GPIO 51 GPIO 52 GPIO 53 GPIO 54 GPIO 55 GPIO 56 GPIO 57 GPIO 58 GPIO 59 GPIO 60 GPIO 61 GPIO 62 GPIO 63 GPIO 64 GPIO 65 GPIO 66 GPIO 67 GPIO 68 GPIO 69 GPIO 70 GPIO 71 GPIO 72 GPIO 73 GPIO 74 GPIO 75 GPIO 76 GPIO 77 GPIO 78 GPIO 79 GPIO 80 GPIO 81 GPIO 82 GPIO 83 GPIO 84 GPIO 85 GPIO 86 GPIO 87 GPIO 88 GPIO 89 GPIO 90 GPIO 91 GPIO 92 GPIO 93 GPIO 94 GPIO 95 GPIO 96 GPIO 97 GPIO 98 GPIO 99 GPIO 100 GPIO	Function assigned to this servo. Setting this to Disabled(0) will setup this output for control by auto missions or MAVLink servo set commands. any other value will enable the corresponding function	✓
SERVO4_MAX	1850	1900	PWM		800 2200	maximum PWM pulse width in microseconds. Typically 1000 is lower limit, 1500 is neutral and 2000 is upper limit.	✓
SERVO4_MIN	1300	1100	PWM		800 2200	minimum PWM pulse width in microseconds. Typically 1000 is lower limit, 1500 is neutral and 2000 is upper limit.	✓
SERVO4_TRIM	1500	1500	PWM		800 2200	Trim PWM pulse width in microseconds. Typically 1000 is lower limit, 1500 is neutral and 2000 is upper limit.	✓
SERVO5_FUNCTION	57	33			-1 GPIO 0 Disabled 1 GPIO 2 GPIO 3 GPIO 4 GPIO 5 GPIO 6 GPIO 7 GPIO 8 GPIO 9 GPIO 10 GPIO 11 GPIO 12 GPIO 13 GPIO 14 GPIO 15 GPIO 16 GPIO 17 GPIO 18 GPIO 19 GPIO 20 GPIO 21 GPIO 22 GPIO 23 GPIO 24 GPIO 25 GPIO 26 GPIO 27 GPIO 28 GPIO 29 GPIO 30 GPIO 31 GPIO 32 GPIO 33 GPIO 34 GPIO 35 GPIO 36 GPIO 37 GPIO 38 GPIO 39 GPIO 40 GPIO 41 GPIO 42 GPIO 43 GPIO 44 GPIO 45 GPIO 46 GPIO 47 GPIO 48 GPIO 49 GPIO 50 GPIO 51 GPIO 52 GPIO 53 GPIO 54 GPIO 55 GPIO 56 GPIO 57 GPIO 58 GPIO 59 GPIO 60 GPIO 61 GPIO 62 GPIO 63 GPIO 64 GPIO 65 GPIO 66 GPIO 67 GPIO 68 GPIO 69 GPIO 70 GPIO 71 GPIO 72 GPIO 73 GPIO 74 GPIO 75 GPIO 76 GPIO 77 GPIO 78 GPIO 79 GPIO 80 GPIO 81 GPIO 82 GPIO 83 GPIO 84 GPIO 85 GPIO 86 GPIO 87 GPIO 88 GPIO 89 GPIO 90 GPIO 91 GPIO 92 GPIO 93 GPIO 94 GPIO 95 GPIO 96 GPIO 97 GPIO 98 GPIO 99 GPIO 100 GPIO	Function assigned to this servo. Setting this to Disabled(0) will setup this output for control by auto missions or MAVLink servo set commands. any other value will enable the corresponding function	✓
SERVO5_MAX	1850	1900	PWM		800 2200	maximum PWM pulse width in microseconds. Typically 1000 is lower limit, 1500 is neutral and 2000 is upper limit.	✓
SERVO5_MIN	1300	1100	PWM		800 2200	minimum PWM pulse width in microseconds. Typically 1000 is lower limit, 1500 is neutral and 2000 is upper limit.	✓
SERVO5_REVERSED	1	0			0 Normal 1 Reversed	Reverse servo operation. Set to 0 for normal operation. Set to 1 to reverse this output channel.	✓
SERVO5_TRIM	1500	1500	PWM		800 2200	Trim PWM pulse width in microseconds. Typically 1000 is lower limit, 1500 is neutral and 2000 is upper limit.	✓
SERVO7_FUNCTION	36	36			-1 GPIO 0 Disabled 1 GPIO 2 GPIO 3 GPIO 4 GPIO 5 GPIO 6 GPIO 7 GPIO 8 GPIO 9 GPIO 10 GPIO 11 GPIO 12 GPIO 13 GPIO 14 GPIO 15 GPIO 16 GPIO 17 GPIO 18 GPIO 19 GPIO 20 GPIO 21 GPIO 22 GPIO 23 GPIO 24 GPIO 25 GPIO 26 GPIO 27 GPIO 28 GPIO 29 GPIO 30 GPIO 31 GPIO 32 GPIO 33 GPIO 34 GPIO 35 GPIO 36 GPIO 37 GPIO 38 GPIO 39 GPIO 40 GPIO 41 GPIO 42 GPIO 43 GPIO 44 GPIO 45 GPIO 46 GPIO 47 GPIO 48 GPIO 49 GPIO 50 GPIO 51 GPIO 52 GPIO 53 GPIO 54 GPIO 55 GPIO 56 GPIO 57 GPIO 58 GPIO 59 GPIO 60 GPIO 61 GPIO 62 GPIO 63 GPIO 64 GPIO 65 GPIO 66 GPIO 67 GPIO 68 GPIO 69 GPIO 70 GPIO 71 GPIO 72 GPIO 73 GPIO 74 GPIO 75 GPIO 76 GPIO 77 GPIO 78 GPIO 79 GPIO 80 GPIO 81 GPIO 82 GPIO 83 GPIO 84 GPIO 85 GPIO 86 GPIO 87 GPIO 88 GPIO 89 GPIO 90 GPIO 91 GPIO 92 GPIO 93 GPIO 94 GPIO 95 GPIO 96 GPIO 97 GPIO 98 GPIO 99 GPIO 100 GPIO	Function assigned to this servo. Setting this to Disabled(0) will setup this output for control by auto missions or MAVLink servo set commands. any other value will enable the corresponding function	✓
THR_FAILSAFE	1	1			0 Disabled 1 Enabled	0 disables the failsafe. 1 enables failsafe on loss of RC input. This is detected either by throttle values below THR_FS_VALUE, loss of receiver valid pulses/data, or by the FS bit in receivers that provide it, like FrSky. If a receiver provides a failsafe signal, it will override this value. If a receiver provides a failsafe signal, it will override this value.	✓

Fig.3.1 Configuration Tab in Mission planar

The configuration tab of the Mission planar Configurator software is specifically set up for the SpeedyBee F405 Wing flight controller, which is running Mission planar firmware version 4.5.2. Mission planar is a popular tool used for configuring and tuning flight controllers in both multirotor and fixed-wing UAVs. In this configuration, the gyro update frequency is set to 8.00 kHz, and the PID loop frequency is set to 4.00 kHz, which ensures responsive and stable flight performance. The accelerometer is enabled to support flight stabilization, while the barometer and magnetometer are disabled, likely for a simplified or manual navigation setup.

The sensor alignment section indicates that the gyro/accelerometer orientation is adjusted to CW 270°, aligning the flight controller's data with its actual mounting orientation inside the aircraft. No manual adjustments have been made to the accelerometer trim, as both pitch and roll values are set to zero. In the DShot Beacon Configuration, the drone is programmed to emit a beeping tone when the receiver signal is lost (RX_LOST) or when a designated AUX switch is activated (RX_SET), which is useful for locating the aircraft in case of emergency landings.

Additionally, under Beeper Configuration, options are enabled to notify the user when the gyro is calibrated and when the radio link is lost, enhancing overall safety. Only the Gyro and Accelerometer sensors are active. The flight controller is currently not armed, and the system reports a CPU load of 30% with a cycle time of 250 μ s, which is within safe operating limits. Overall, this configuration establishes a reliable and stable foundation for fixed-wing or hybrid VTOL aircraft operation using Mission planar.

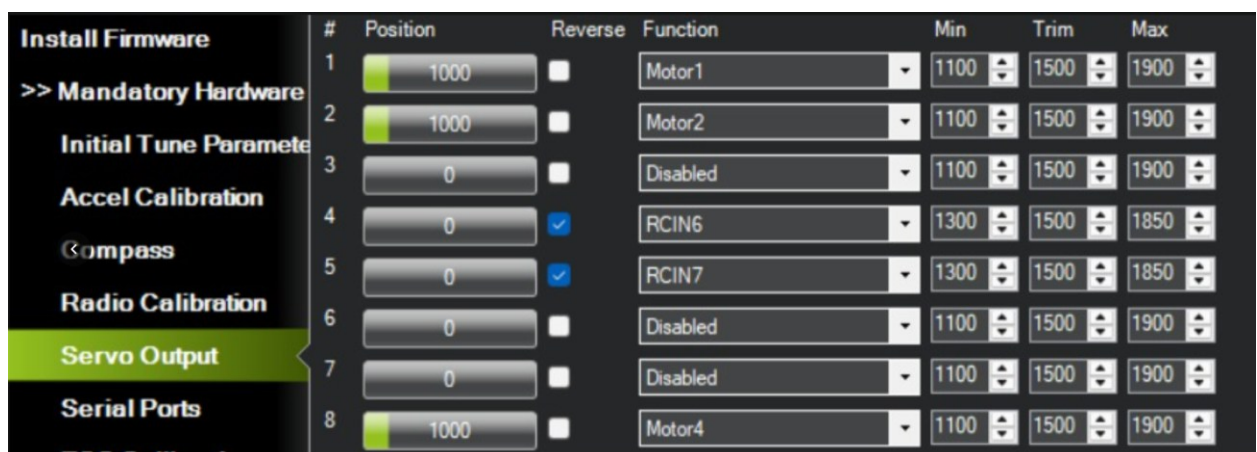


Fig.3.2 Motors Tab in Mission planar

The Motors tab of the Mission planar Configurator software is configured for a SpeedyBee F405 Wing flight controller running Mission planar firmware version 4.5.2. The motors tab is essential for testing, configuring, and verifying motor direction, mixer settings, and ESC (Electronic Speed Controller) behavior during drone or fixed-wing aircraft setup.

In the mixer section, the aircraft is set to the “FLYING WING” configuration. This setup is used for fixed-wing drones that do not have a traditional tail and rely on elevon mixing for pitch and roll control. The mixer layout shows the assigned motor positions and directions. There is also a toggle switch for motor direction reversal, which is useful when matching motor rotation to the propeller type or airframe layout.

The ESC/Motor Features section shows the DSHOT600 protocol selected for motor control, which is a fast and reliable digital ESC communication method. Additional options like MOTOR_STOP, ESC_SENSOR, and bidirectional DShot are shown but not activated. The motor idle speed is set to 5.5%, which ensures a stable spin at minimum throttle.

On the right, the live sensor data graph shows real-time orientation changes in the X, Y, and Z axes. The voltage and amperage readings are near zero, indicating the system is powered only via USB and the motors are inactive. The motor test sliders at the bottom allow for manual testing of each motor, which is essential for confirming correct motor spin direction and position before flight.

A warning is also displayed reminding users to remove all propellers before testing to avoid serious injury. System stats such as cycle time, CPU load, and configurator version are also shown, helping assess performance and stability.

3.2 TRANSCEIVER SETUP FOR RC PLANE

The Flysky FS-i6S 2.4GHz 10-channel transmitter, paired with the FS-iA10B 10-channel receiver, forms a versatile and dependable radio communication system widely utilized in RC plane and drone applications. In the context of an RC fixed-wing aircraft, this setup ensures interference-resistant and stable communication through AFHDS 2A technology, allowing precise control over essential flight functions such as throttle, elevator, ailerons, rudder, and auxiliary channels for components like flaps or camera gimbals. The FS-i6S transmitter is equipped with a modern touchscreen interface, an ergonomic design, and telemetry capabilities, making it suitable

for both novice and experienced users. Meanwhile, the FS-iA10B receiver supports both PWM and PPM output formats, enabling seamless integration with popular flight controllers such as the SpeedyBee F405 Wing. With support for up to ten configurable channels, this system provides the flexibility required for advanced flight modes, VTOL transition management, and real-time in-flight adjustments, making it a reliable and scalable solution for fixed-wing UAV platforms.



Fig.3.3 The Flysky FS-i6S 2.4GHz 10-Channel Transmitter

Channel 1, 3, 7, and 8 were used for the RC plane. Channels 3, 7, and 8 are assigned to servo motors and channel 1 is assigned to the BLDC motor.

4.RESULTS

4.1 FLIGHT TESTING

The flight test was conducted with model built based on the same design with a few modifications.



Fig.4.1 full Model of the Plane Model

There were issues with the model such as weight balancing, and the material losing its rigidity after being cut due to air moisture (which necessitated the application of tape). These issues were only discovered during the test flight. it was difficult for the plane to produce initial lift for takeoff. Prior to the test, a simple thrust check was conducted.

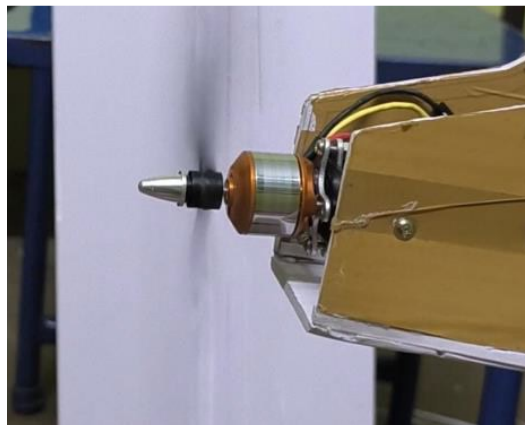


Fig.4.2 Mock Run to Check Thrust

CHAPTER 5

CONCLUSION

The project was successfully completed as a functional prototype, demonstrating the feasibility of a fixed-wing UAV platform for surveillance applications. The prototype validated key design decisions in electronics integration, flight controller configuration, and RC communication, serving as a strong proof-of-concept for future development. The team intends to continue building upon this foundation with further refinements in subsequent iterations.

While the prototype achieved its primary objectives of hardware assembly, sensor calibration, and ground testing, certain airframe limitations were identified during the process. The airframe was constructed entirely from scratch through a structured process of trial and error, which itself yielded significant engineering insight. The challenges observed and areas identified for improvement are as follows:

- Heavy weight of the frame reducing the overall thrust-to-weight ratio
- Weight imbalance affecting directional stability during ground runs
- Motor KV rating not optimally matched to the propeller and airframe weight, resulting in lower-than-required thrust
- Wing profile not conforming to a proper aerofoil shape, limiting lift generation

Key Learnings:

- Interfacing and configuring a Flight Controller with firmware tools
- Calibration of Flight Controller sensors including gyroscope and accelerometer
- 3D printing techniques and associated structural considerations
- Fixed-wing drone aerodynamics and associated technologies
- Operation and channel mapping of a radio controller
- Prototyping and iterative design using foam boards

Future Improvements:

- Redesigning the airframe with lighter material or an optimized 3D-printed structure to improve the thrust-to-weight ratio
- Incorporating a properly profiled aerofoil wing to maximize aerodynamic lift
- Selecting a lower KV motor paired with a larger propeller better suited to the airframe weight
- Precise centre-of-gravity balancing prior to flight testing
- Integration of a GPS module and telemetry system for autonomous waypoint navigation
- Addition of a camera payload for live surveillance feed during flight

This project provided invaluable hands-on experience in fixed-wing UAV development, embedded systems integration, and the practical realities of drone airframe design. The successful assembly and ground testing of a functional prototype is a significant milestone. The knowledge and skills gained during this internship will serve as a strong foundation for future work in autonomous aerial systems and drone engineering, with the next phase aimed at achieving a complete and stable flight.